Arpit Singh

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Education

University of Surrey, Guildford — Guildford, UK

MSc Business Analytics – Distinction — *Sept 2025*

- Key Modules: Operational Analytics (83%), Financial Modelling (80%), Statistics & Econometrics (77%), Data Mining (77%), Machine Learning, AI & Visualisations (75%)

Experience

HSBC UK

Universal Banker – WPB Data & Analytics — *Mar 2026 – Present*

- Investigate and resolve 2–3 financial crime cases daily, analysing customer account activity and transaction patterns across multiple data points to identify fraud indicators and escalate high-risk cases within an FCA-regulated environment.
- Built an end-to-end financial operations analytics project on 5,000 customers and £28.5M revenue data, delivering a churn prediction model identifying 652 at-risk customers, 12-month revenue forecast of £12.5M at 0.73% MAPE, and RFM segmentation representing £1.6M annual revenue at risk.
- Collaborate with internal Risk, Fraud, and Business teams to share case findings, escalate high-risk patterns, and contribute to process improvements across financial crime prevention frameworks.

Industry Pro

Data Consultant – Remote — *Sept 2024 – Feb 2026*

- Built a provider scoring model ranking 150+ verified service providers across response time, job success rate, and service category coverage, directly informing provider allocation decisions for enterprise clients including BHEL and Parle Agro across VFD, PLC, and HMI repair categories.
- Developed cross-market reporting consolidating India and US service operations following the company's 2025 expansion, reducing senior leadership reporting time from 3 days to same-day delivery across 6 service lines.
- Analysed energy savings ROI across client sites, quantifying automation upgrade outcomes to support the sales team in client retention and new business pitches.

Junior Data Analyst — *Aug 2023 – Aug 2024*

- Built and maintained operational trackers monitoring 500+ monthly service requests across VFD, PLC, HMI, and control panel categories, surfacing provider performance trends and job completion rates for the founding team.
- Identified a 31% gap in response times between top and bottom quartile providers by analysing service request data, directly informing the founding team's provider quality review process.
- Cleaned and standardised data across 150+ onboarded service providers, improving data completeness from 64% to 91% across 12 key operational fields and supporting early resource allocation decisions.

Projects

Financial Operations Analytics | *Python, Pandas, Scikit-learn, Prophet, Statsmodels*

- Built a churn prediction model across 5,000 customers identifying 652 at-risk customers, with recency and transaction frequency as dominant predictors and outputs translated into an executive dashboard.
- Developed an ARIMA revenue forecasting model on 5 years of synthetic banking data aligned to FCA reporting conventions, projecting 12-month revenue at 0.73% MAPE and detecting Q4 seasonal peaks to support capacity and resource planning.
- Delivered RFM segmentation across 8 customer groups and CLV analysis revealing a 14.33x CLV-to-CAC ratio and 5-month average customer payback period, informing targeted retention strategy across 4,344 active accounts.

Retail Sales Analytics & Forecasting | *SQL, Python, Power BI, Prophet, DAX*

- Executed end-to-end SQL diagnostics on 9,800+ retail transactions worth $2.26M across 4 regional markets, applying window functions and CTEs to identify a 32% sales collapse in the South region and low-AOV structural weaknesses in the Central market.
- Built a three-page interactive Power BI dashboard with DAX measures, time intelligence modelling, and KPI tracking – enabling regional performance comparison and data-driven business recommendations.

- Developed a Python Prophet forecasting model reducing MAPE from ∼250% to ∼11% by aggregating daily transaction data to weekly time series, improving forecast reliability and automating demand planning.

Impact of Statistical Visualisations on Decision-Making | *R, MATLAB, SQL, Excel, iMotions*

- Designed and executed a controlled behavioural analytics study across 113 participants using eye-tracking, pupil dilation analysis, and physiological sensors to measure how visualisation design affects decision accuracy and cognitive load.
- Applied Wilcoxon testing and linear mixed-effects modelling in R across multimodal datasets combining response accuracy, fixation heatmaps, and biometric signals.
- Found that redesigned violin plots improved interpretation accuracy from 61% to 70% (p=0.014) and reduced response times by 14 seconds compared with traditional histograms.

Road Traffic Accident Severity Prediction | *SAS, Machine Learning, NLP*

- Conducted end-to-end predictive analytics on 2,480 UK road traffic accident records across 35 variables using SAS for data preparation, visualisation, and modelling.
- Developed a Gradient Boosting model achieving 83.47% accuracy, 0.9172 AUC, and 0.6688 KS statistic, identifying key accident severity drivers including location and speed limit.
- Applied SAS text analytics and topic modelling on 598 social media posts to generate insights on accident impact and public sentiment.

Technical Skills

Languages & Tools: SQL (MySQL), Python (Pandas, NumPy, Prophet), R, SAS, STATA, MATLAB
BI & Visualisation: Power BI, Power Query, DAX, Excel (Advanced), Data Visualisation, Dashboarding
Analytics: Statistical Modelling, Forecasting, Machine Learning, Regression Analysis, ETL, Behavioural Analytics, Exploratory Data Analysis (EDA)

Certifications

Databricks Generative AI Fundamentals · HSBC Data Analytics Foundation